

37 Surplus & surplus management

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2026 objectives: 5.1.1, 5.2.1, 5.2.3-5.2.6

5 related questions are filed elsewhere in this volume. Find them by selecting this chapter and enabling related topics in [index.html](#).

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CA1-P1 April 2007 Q4	15 marks
CA1-P1 April 2009 Q3	17 marks
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CA1-P1 April 2013 Q4	12 marks
CA1-P1 September 2014 Q7	22 marks
CA1-P1 September 2015 Q6	22 marks
CA1-P1 April 2016 Q4	12 marks
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CA1-P1 April 2017 Q4	20 marks
CP1-P1 September 2019 Q4	11 marks
CP1-P1 April 2021 Q2	7 marks
CP1-P1 September 2022 Q3	5 marks

CA1-P1 | April 2007 | Q4

Current topic | 15 marks total | Original PDF p3

Examiner report and original paper: use the links for this question in [index.html](#).

- 4** The three yearly valuation of a defined benefit scheme has just been completed. A large surplus has emerged, which has arisen from two sources:
- A large increase in the market value of the scheme's equity investments during the period since the last valuation, considerably in excess of the assumption used in the previous valuation.
 - A reduction in the active membership a year ago, when the sponsoring employer reorganised the business and made a reduction in staff numbers. The benefits for early leavers in the scheme are such that significant withdrawal surpluses arise.

These two features have had approximately equal impact.

The scheme's actuary has advised that two changes can be made without detriment to the scheme's long term financial position:

- The employer's contribution rate can be halved from 8% to 4% of scheme salaries for each of the next three years.
- The rate of pension increases in payment can be changed from a fixed 3% per annum to the rate of inflation. This change to the pension increase would apply to pensioners and deferred pensioners as well as current contributing members.

These two changes have the same expected cost, and implementing both of them is affordable.

The employer has proposed that it will implement the contribution reduction, but that there will be no changes to the pension increases. The balance of the surplus will be retained in the scheme.

Discuss this proposal for using the surplus. [15]

You need not assume that the scheme is written under UK legislation, or that of any other state.

CA1-P1 | April 2009 | Q3

Current topic | 17 marks total | Original PDF p2

Related: Ch 17.

3 A life insurance company launched a single premium unit-linked bond product five years ago. The terms of the contract were as follows:

- Annual Management Charge of 1% of the accumulated fund each year
- early exit charges to apply in first year at 3% of the accumulated fund, second year at 2% and in third year at 1%
- guaranteed return of premium on death

The policyholders have the choice of three funds to invest in:

- an equity fund
- a fixed interest fund
- a fund that guarantees at least the return of premium on every anniversary of the policy

(i) Describe the approach that the company would have used to project its expected profits when it launched the product. [6]

Having reviewed the policies' actual performance it is seen that the product has made losses every year since its launch.

(ii) Discuss reasons why the losses may be occurring. [7]

(iii) Outline the actions that management can take to reduce the losses on this product. [4]

[Total 17]

CA1-P1 | April 2010 | Q6

Current topic | 32 marks total | Original PDF p3

Related: Ch 18, Ch 35.

- 6** A mutual life insurance company has a large number of policyholders and a significant undistributed surplus. This undistributed surplus has built up over many years.
- (i) Outline how the undistributed surplus may have arisen. [5]
 - (ii) List the data required to calculate the value of the provisions needed for the in-force policies. [4]
 - (iii) Outline why the data used to determine these provisions would be grouped. [2]
 - (iv) Explain the considerations that should be taken into account when validating the grouping of the data. [4]
- The company intends to demutualise and become a limited liability company. On demutualisation, the undistributed surplus is to be distributed to the policyholders in the form of both additional benefits and shares in the limited company. The distributed shares will provide the company with capital.
- (v) Explain why the company needs to retain some of the surplus as capital. [6]
 - (vi) Discuss the considerations to be taken into account in determining the split between the additional benefits and the shares. [6]
 - (vii) Discuss how the distribution of future surplus would change following the demutualisation. [5]
- [Total 32]

CA1-P1 | September 2012 | Q2

[Current topic](#) | [9 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 2** A large defined benefit pension scheme has made a significant profit from selling fine art assets that were part of its investment portfolio and now has a substantial surplus.
- (i) Outline the options available for using this surplus. [3]
- (ii) Discuss the factors that will influence how and when the surplus is used by the scheme. [6]
- [Total 9]

CA1-P1 | April 2013 | Q4

Current topic | 12 marks total | Original PDF p2

Related: Ch 17.

- 4** A large insurance company is reviewing the profitability of its pet insurance product. The pet insurance pays out a fixed amount of money on the death of the insured pet and will also pay 75% of any veterinary bills as long as the treatment provided is on the approved list set out in the policy document. A number of exclusions and restrictions apply.
- (i) Outline how to project the expected profits from this product. [7]
- (ii) Outline factors which might lead to the actual profits for this product being worse than expected. [5]
- [Total 12]

CA1-P1 | September 2014 | Q7

Current topic | 22 marks total | Original PDF p4

Related: Ch 31.

7 A large insurance company offers a with profits product with the following features:

- The term is 10 years.
- On withdrawal before the end of the 10 year term, the surrender value is the accumulated fund value with bonuses. There are no withdrawal penalties.
- At the end of 10 years, the maturity value is the higher of the accumulated fund value with bonuses or the original investment accumulated at 3% a year.
- The stated aim is to provide bonuses such that the total investment return will be between 1% and 3% a year in excess of the return on an established domestic corporate bond index over the term of 10 years.

The insurance company invests the premiums in corporate bonds, held equally in the domestic market and emerging foreign markets. It hopes that the higher expected return from emerging foreign markets will help to fund the bonuses required to meet the product's stated aim.

(i) Explain why the valuation for internal management purposes may use different assumptions to those used for a statutory solvency valuation. [4]

(ii) Describe how any surplus may be distributed. [7]

The insurance company intends to distribute the majority of its surplus as an annual bonus with an additional terminal bonus for investors who keep their investments for the full 10 year term. It expects that for a typical policyholder investing in the product for the full 10 year term approximately 80% of the distributed surplus will be from annual bonuses and 20% from terminal bonuses.

Returns on the domestic corporate bond index have consistently been between 5% and 7% a year. There have been recent concerns regarding the domestic economy and some analysts are suggesting that returns in the medium term may be lower, possibly between 2% and 5% a year.

(iii) Discuss the impact this revised economic outlook may have on the bonus distribution policy. [5]

One of the insurance company's directors has suggested that, with the possibility of falling bond yields, the proportion of surplus distributed by annual bonuses should reduce to 50%. He has also suggested introducing withdrawal penalties.

(iv) Outline the advantages and disadvantages of these two suggestions. [6]
[Total 22]

CA1-P1 | September 2015 | Q6

Current topic | 22 marks total | Original PDF p3

Related: Ch 35.

6 (i) List the reasons why a health insurance company might want to analyse its financial results. [5]

(ii) Outline possible sources of surplus for a health insurance company. [4]

A health insurance company has made losses over the last few years and its solvency position has deteriorated.

(iii) Explain the actions that it could take to reduce the deterioration in this company's solvency position. [13]

[Total 22]

CA1-P1 | April 2016 | Q4

[Current topic](#) | [12 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 4** A life insurance company has conducted an analysis of surplus that shows that there is a significant source of surplus arising from surrenders.

- (i) Outline the principal reasons why the company may have experienced such high surrender profits. [4]

The marketing manager has suggested that the company could increase bonuses to distribute this surplus.

- (ii) Comment on this proposal for using the surplus. [5]

- (iii) Suggest alternative uses of the surplus. [3]

[Total 12]

CA1-P1 | April 2017 | Q2

[Current topic](#) | [8 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 2** A proprietary life insurance company was formed two years ago and does not write any with profits business. During its second financial year it underwent a very rapid increase in the volumes of new business written. It has just analysed the change in its surplus for the second financial year.

Discuss why the Directors are interested in that analysis. [8]

CA1-P1 | April 2017 | Q4

Current topic | 20 marks total | Original PDF p2

Related: Ch 7, Ch 17.

4 A general insurance company provides liability insurance for hairdressers.

- (i) Outline four types of liability that may be covered. [4]

The number of policies and premium income for this business has been increasing each year. The financial results for this business, however, show that profits are reducing each year, compared to what was expected.

- (ii) Describe how the expected profits for this business could be projected in future years. [4]

- (iii) Outline possible reasons for the reduction in profitability. [5]

- (iv) Suggest levers that the company could use to increase profits. [3]

- (v) Suggest reasons why the insurance company may wish to continue to offer this insurance. [4]

[Total 20]

CP1-P1 | September 2019 | Q4

[Current topic](#) | [11 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 4** Discuss why an insurance company may not pay a terminal bonus on one or more of its with-profits products. [11]

CP1-P1 | April 2021 | Q2

Current topic | 7 marks total | Original PDF p2

Related: Ch 35.

- 2** A general insurance company's latest annual profits show that the company is performing in line with expectations overall. However, the performance of its home insurance product has been poor over the last 12 months, and it has suffered significant losses on this business.

- (i) Suggest possible reasons for the losses on the home insurance product. [4]

The regulator has reviewed the general insurance company's reserves and has determined that they are not sufficient. The regulator has therefore instructed the insurance company to increase its reserves within 1 month of the date of the instruction.

- (ii) Outline the options available to the insurance company to increase its reserves. [3]

[Total 7]

CP1-P1 | September 2022 | Q3

[Current topic](#) | [5 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 3** A defined benefit scheme that is open to new entrants and future accrual has shown in its latest accounts that the scheme's funding deficit has deteriorated since the last scheme valuation.

Suggest possible reasons that could explain the change in funding deficit shown in the accounts. [5]